

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Conduct of Operations		
TASK:	Calculate Shutdown Margin		
TASK NUMBER:	N1200030301		
JPM NUMBER:	11-01 NRC RO Admin A1-1		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:		Classroom	
REFERENCES:	SC.RE-ST.ZZ-0002(Q), Rev. 20 SHUTDOWN MARGIN CALCULATION S1.RE-RA.ZZ-0016(Q), Rev. 4 CURVE BOOK		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	30 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	40 minutes		
Developed By:	G Gauding Instructor	Date:	8-14-12
Validated By:	Mulford/Brennan SME or Instructor	Date:	8-24-12
Approved By:	Training Department	Date:	
Approved By:	Operations Department	Date:	
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

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NAME: _____

DATE: _____

{PRIVATE }SYSTEM: Conduct of Operations

TASK: Calculate Shutdown Margin

TASK NUMBER: N1200030301

INITIAL CONDITIONS: Unit 1 Reactor Power is 75%. Control Bank D is at 170 steps.
Current boron concentration is 100 ppm.
Current core burnup is 11,890 EFPH.

INITIATING CUE: Control Rod 1D5 has been declared INOPERABLE at 1000 today due to not moving during a rod exercise test. You have been directed to perform a Shutdown Margin Calculation IAW SC.RE-ST.ZZ-0002 to satisfy the action requirement of TSAS 3.1.3.1 Action c.3

The current time is 1020.

This is a time critical JPM. Time will commence when you are provided with the applicable procedures.

{PRIVATE }Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Perform SDM calculation and determine SDM is SAT, and within a band of -2513 to -2646 pcm.

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System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Obtains copy of procedure SC.RE-ST.ZZ-0002(Q). Obtains copy of S1.RE-RA.ZZ-0016 Curve Book.	Provide Rev. 20 of surveillance. Provide Rev. 4 of Curve Book. Evaluator: MARK start time of JPM: _____		
	3.0	Reviews and signs off Precautions and Limitations Section 3.0	Operator reviews Precautions & Limitations and signs off Steps 3.1 – 3.11.		
	5.1	SELECT the applicable step below (e.g. 5.1.1.A):	Operator determines 5.1.1.C is the correct step for the current conditions, which is to perform Attachment 3.		
	5.1.1.C	<u>IF</u> SDM is to be verified (per T/S 4.1.1.1.a) with reactor critical (Mode 1 or 2) and one or more control rods are inoperable, <u>THEN</u> COMPLETE Attachment 3.	Operator refers to Attachment 3.		
	Att. 3 2.1	The reactor is in Mode 1 or Mode 2 with $k_{eff} \geq 1.0$	Operator determines Unit is in Mode 1.		
	Att. 3 3.0	Precautions and Limitations: Section 3.0	Operator reviews and signs steps 3.1-3.4.		

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System: Conduct of Operations**Task:** Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 3 Section 4	Section 4 contains the data points required and the calculations to determine shutdown margin.	Operator determines the following data and calculations: Allowable ranges are shown, exact numbers using Tables and adjusting for 75% power is in parentheses. 4.1.1 75% 4.1.2 100 ppm 4.1.3 D @ 170 steps 4.1.4 11,890 EFPH 4.1.5 1 4.2.1 -3450- -3460 pcm (-3454) 4.2.2 -3908- -3918 pcm (-3913) 4.2.3 830-835 pcm (832) 4.2.4 830-835 pcm (832) 4.2.5 300-340 pcm (325.7) 4.2.6 -5368- -5398 pcm (-5377.3) 4.3.1 -5368 - -5398 pcm (5377.3) 4.3.2 652.8-654.3 pcm (653.5) 4.3.3 2100-2200 pcm (2126) 4.3.4 -2513.7- -2645.2 pcm (-2597.8)		

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System: Conduct of Operations**Task:** Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	4.4	Att 4 Acceptance Criteria 4.4.2 IS the SDM (Item 4.3.4) EQUAL TO OR MORE NEGATIVE THAN (-)1300 PCM?	Operator reviews required SDM in MODE 1 or 2 of -1300 pcm and initials 4.4.1. Operator determines calculated SDM is SAT and initials SAT.		
*			Evaluator: MARK stop time of JPM: _____ Total time from START to STOP is 40 minutes or less.		

Terminating Cue: Once operator has returned procedures to proctor, terminate JPM.

Note to Evaluators: The bands for acceptable pcm were derived using the graphs provided in S1.RE-RA.ZZ-0016 Curve Book. The SDM calculation allows using Graphs vs. Tables in some steps, where the Tables have a number value and a Graph is open to extrapolation.

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JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- _____ 1. Task description and number, JPM description and number are identified.
- _____ 2. Knowledge and Abilities (K/A) references are included.
- _____ 3. Performance location specified. (in-plant, control room, or simulator)
- _____ 4. Initial setup conditions are identified.
- _____ 5. Initiating and terminating Cues are properly identified.
- _____ 6. Task standards identified and verified by SME review.
- _____ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- _____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
- _____ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

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